

SML Assignment 1

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Code Overview

Preprocessing

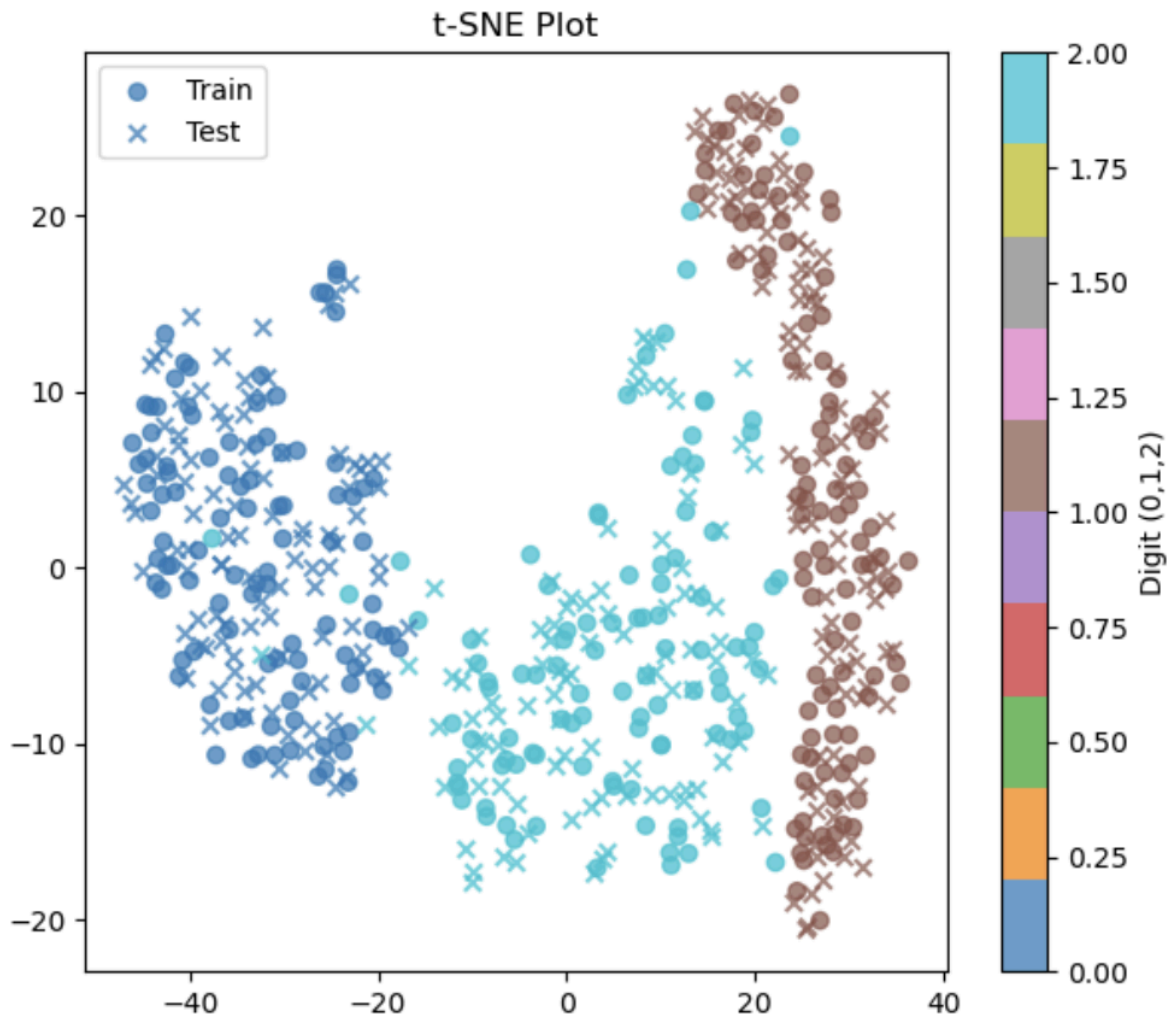
1. Read the images and labels into numpy arrays, and filter for 0,1, and 2.
2. Reshape images into a stacked feature vector using `np.reshape()` (28x28 to 784x1) and then dividing by 255 for normalisation

MLE, LDA & QDA

For each class:- zero, one, and two

1. Compute MLE mean, this is an average over the sample data points, so `sum(samples)/len(samples)`.
2. Compute MLE covariance matrix, using the formula given in the assignment, we computed sample multiply sample transpose which is a 784x784 matrix for each sample and then averaging .
3. `np.linalg.det(covariance matrix)` reveals the matrix is singular, so to calculate the inverse we add a small value epsilon times identity matrix to the covariance matrix as discussed in lecture slides.
4. `np.array_equal()` reveals that covariance matrices for classes zero,one, and two are not equal (case 3 not case 2), and since they were singular to begin with, they are obviously not a scaling of the identity matrix.
5. Hence, QDA was chosen, as per the 3 cases for MVG discriminant analysis discussed in lecture slides as well as the Duda book.
6. Due to ambiguity in the assignment (LDA **or** QDA mentioned first, but every subsequent mention and assignment description mentions LDA **and** QDA), I also implemented an LDA classification, though it predictably had a lower accuracy as opposed to the QDA classification, here I used an average of the covariance matrices of the three classes as the common covariance matrix, which makes some sense to me as it is the expected covariance matrix for all of these classes, assuming they have equal priors.
7. Which brings us to the next point, the implementation assumes equal priors for all classes.
8. Parameters for LDA and QDA were precomputed to avoid repeated computation for each classification and then passed to the discriminant functions, argmax of the discriminant values of the point was chosen as the final class for both LDA and QDA.
9. Linear and quadratic discriminants were computed using the standard formulae for their corresponding quadratic, linear and constant terms (by taking natural log(posterior) and then removing the terms which are constant irrespective of the class i.e. constants like 2, pi etc. and terms in linear discriminant which are independent of mean and sample point, as these constants do not affect our final decision which is based on comparison).
10. For `np.random.seed(14)`, QDA (0.99 or 99%) had greater accuracy than LDA (0.9 or 90%)

t-SNE Plot



Dark blue corresponds to Zero

Light blue corresponds to Two

Brown corresponds to One

Zero and Two are clustered almost uniformly around their means and seem to have a similar covariance matrix, while One has significantly more variance along one principal component than the other out of the two taken for this plot.

Train and Test points both roughly follow similar distributions for the classes, with a few points being significantly farther away from the mean, sometimes inside the cluster of points corresponding to the other 2 classes.